

Supporting information

Soldering of iron catalysts for direct synthesis of light olefins from syngas under mild reaction conditions

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Table S1. XRF analysis of metal content in the catalysts before and after catalysis

Samples	Size of Fe₂O₃ and Fe carbide, nm (XRD)	Fe, wt %	Bi, wt %	Pb, wt %
Fe/SiO ₂ -before	17	15.46	-	-
Fe/SiO ₂ -after	4.4	14.48	-	-
FeBi/SiO ₂ -before	15	14.43	0.94	-
FeBi/SiO ₂ -after	5.8	16.23	1.10	-
FePb/SiO ₂ -before	16	14.5	-	1.27
FePb/SiO ₂ -after	5.2	13.8	-	1.25

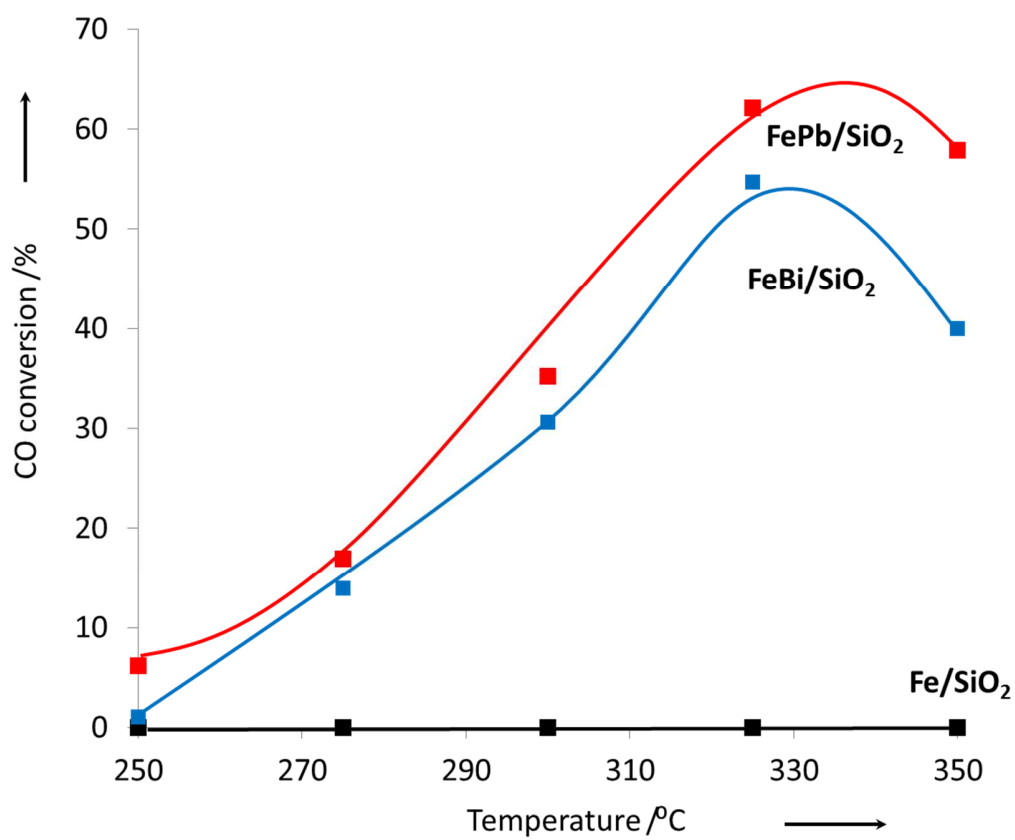


Figure S1. Effect of the type of promoter on the CO conversion at different temperatures. Reaction conditions: P= 5 bar, H₂/CO=1, GHSV =3.4 Lg⁻¹h⁻¹.

Comment: The catalytic results were obtained by decreasing temperature in the row 350-325-300-275-250°C.

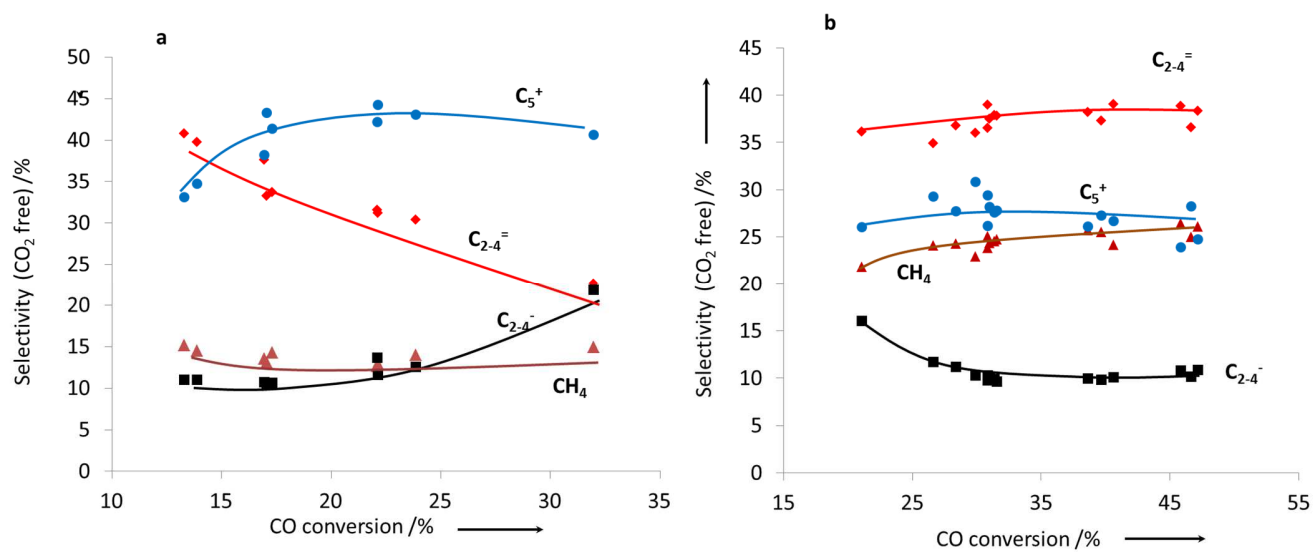


Figure S2. Selectivity to hydrocarbons versus CO conversion over Fe/SiO₂ (a) and FeBi/SiO₂ (b) catalyst. Reaction conditions: P= 10 bar, H₂/CO=1, GHSV =1.5-27 Lg⁻¹h⁻¹, T=350 °C.

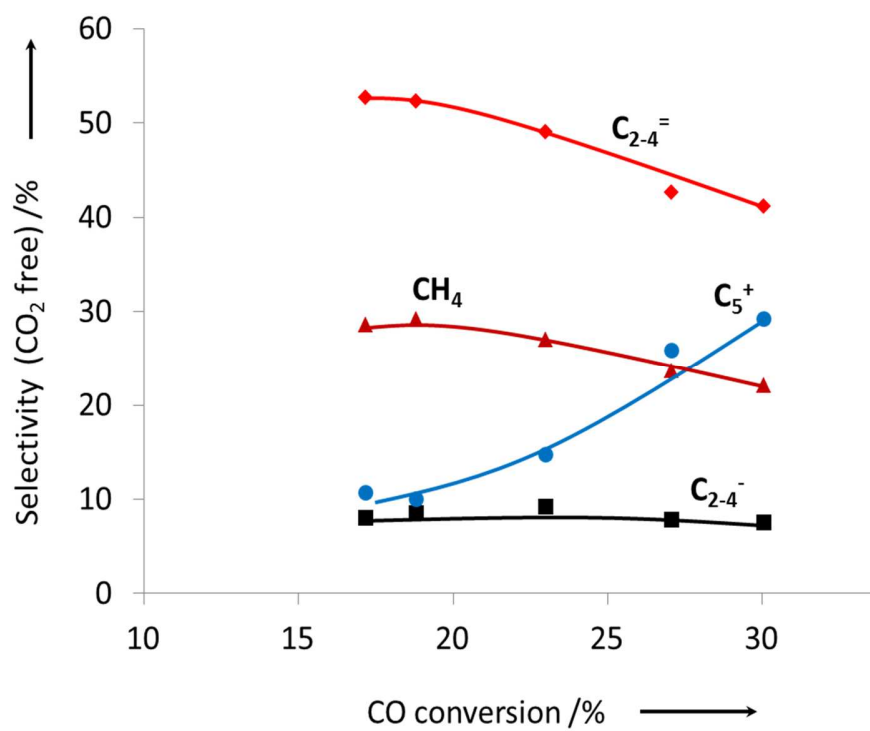


Figure S3. Selectivity to hydrocarbons versus CO conversion over the FeBi/SiO₂ catalyst. Reaction conditions: P= 1 bar, H₂/CO=1, GHSV =1.5-7.5 Lg⁻¹h⁻¹, T=350 °C.

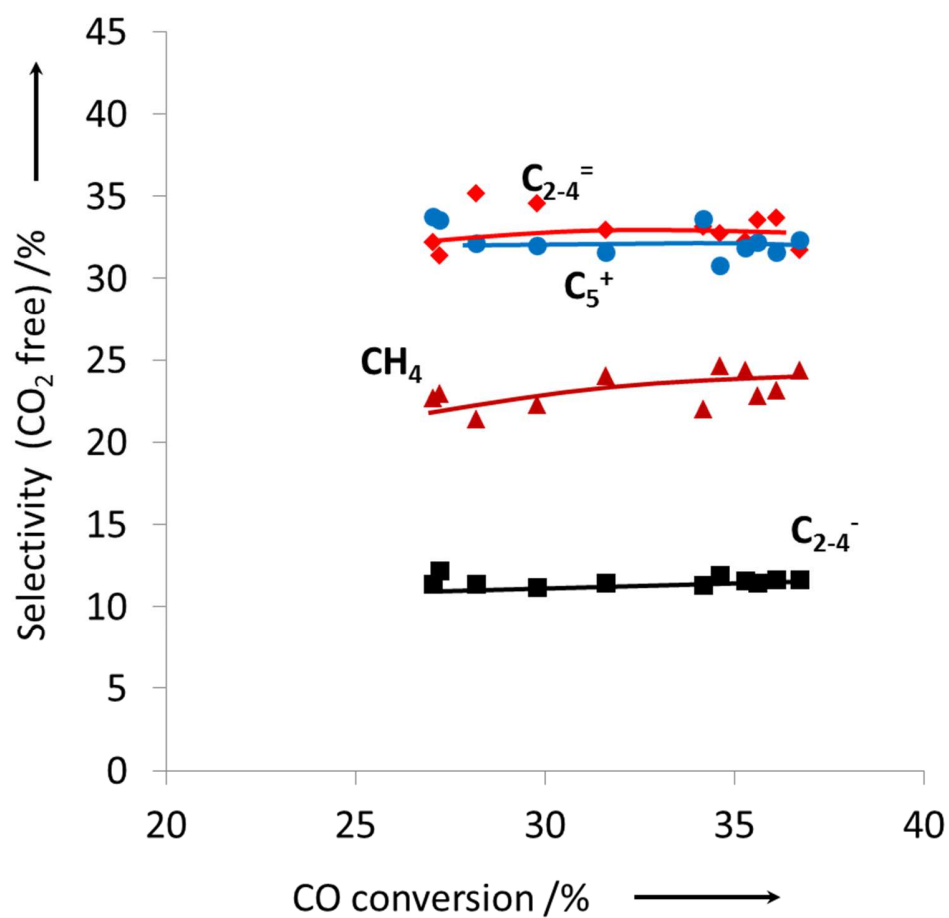


Figure S4. Selectivity to hydrocarbons versus CO conversion over FePb/SiO₂ catalyst. Reaction conditions: P= 10 bar, H₂/CO=1, GHSV =1.5-27 Lg⁻¹h⁻¹, T=350 °C

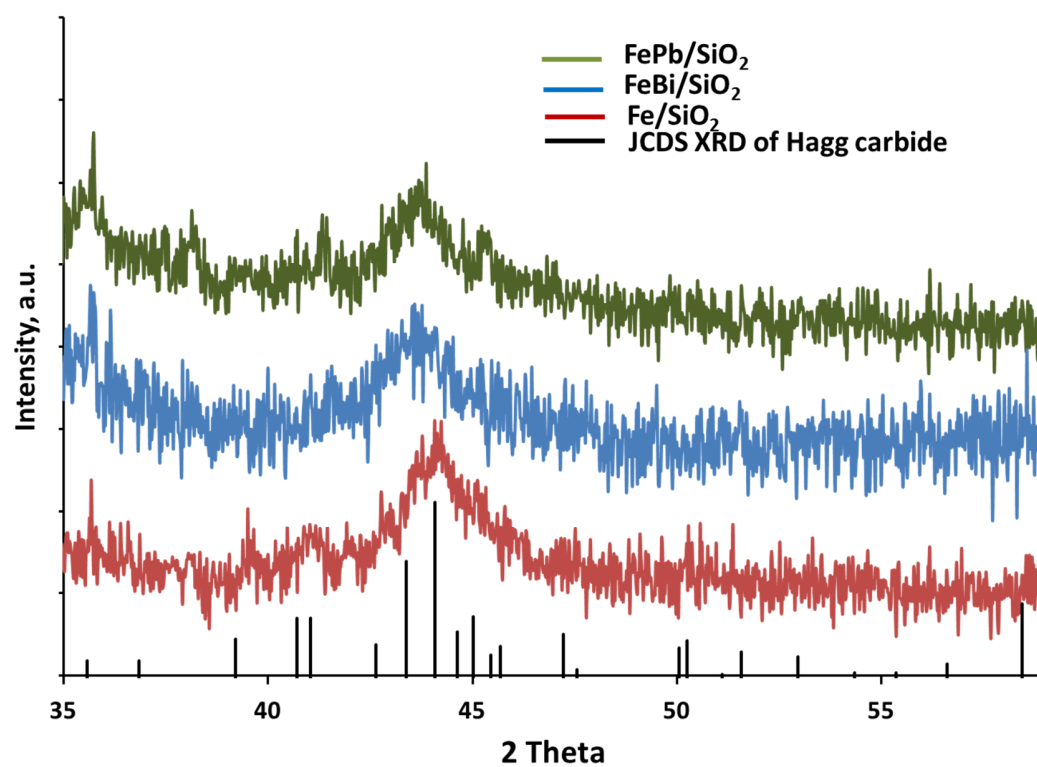


Figure S5. XRD patterns of the catalysts after FT reaction

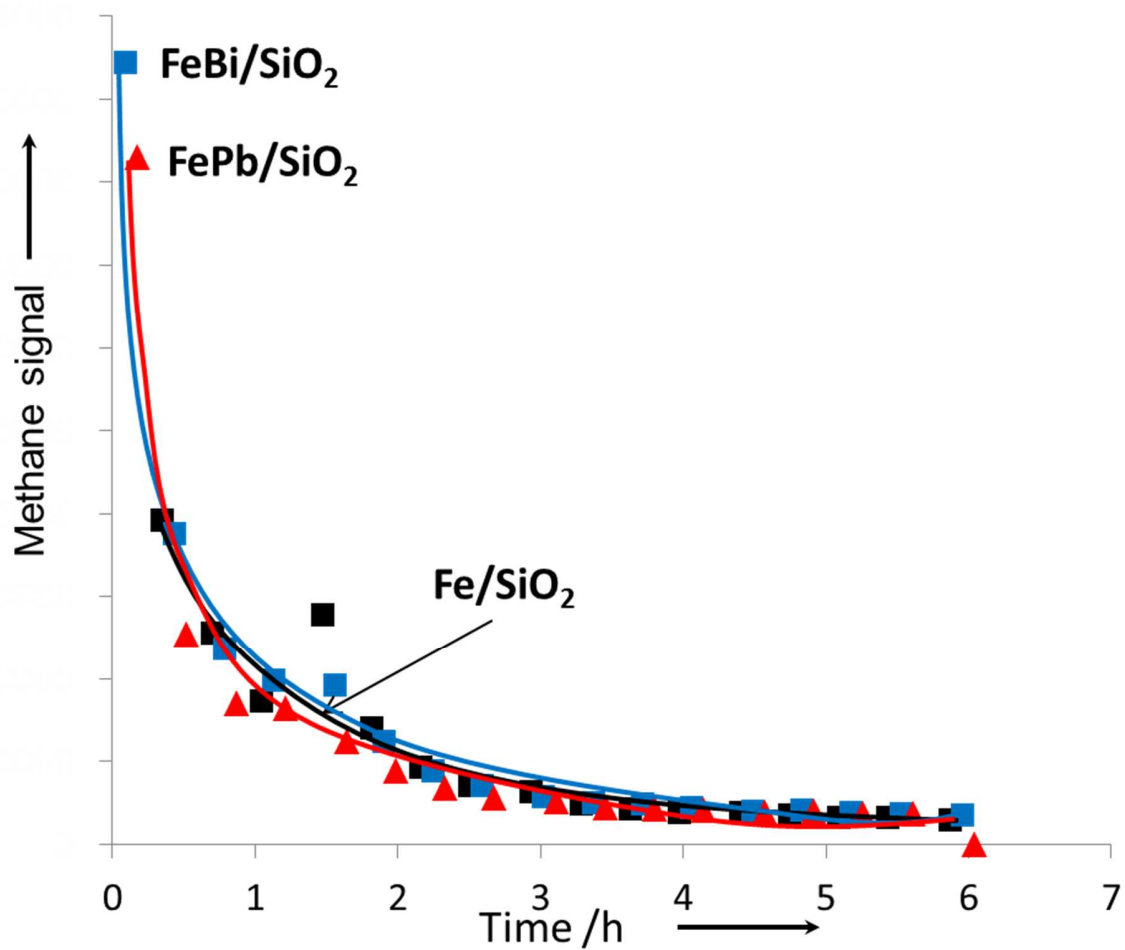


Figure S6. Methane formation rate in static hydrogenation of carbidized Fe/SiO₂, FePb/SiO₂ and FeBi/SiO₂

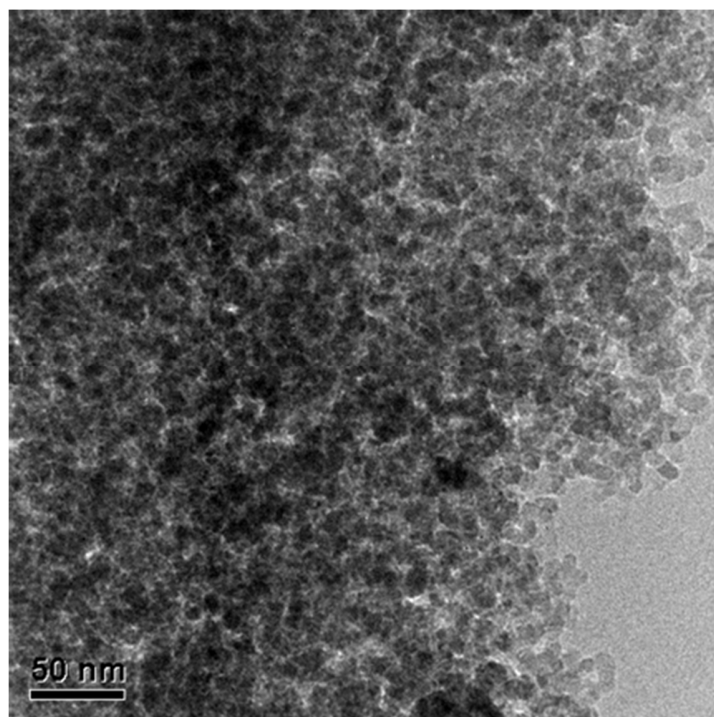


Figure S7. TEM image of Fe/SiO₂

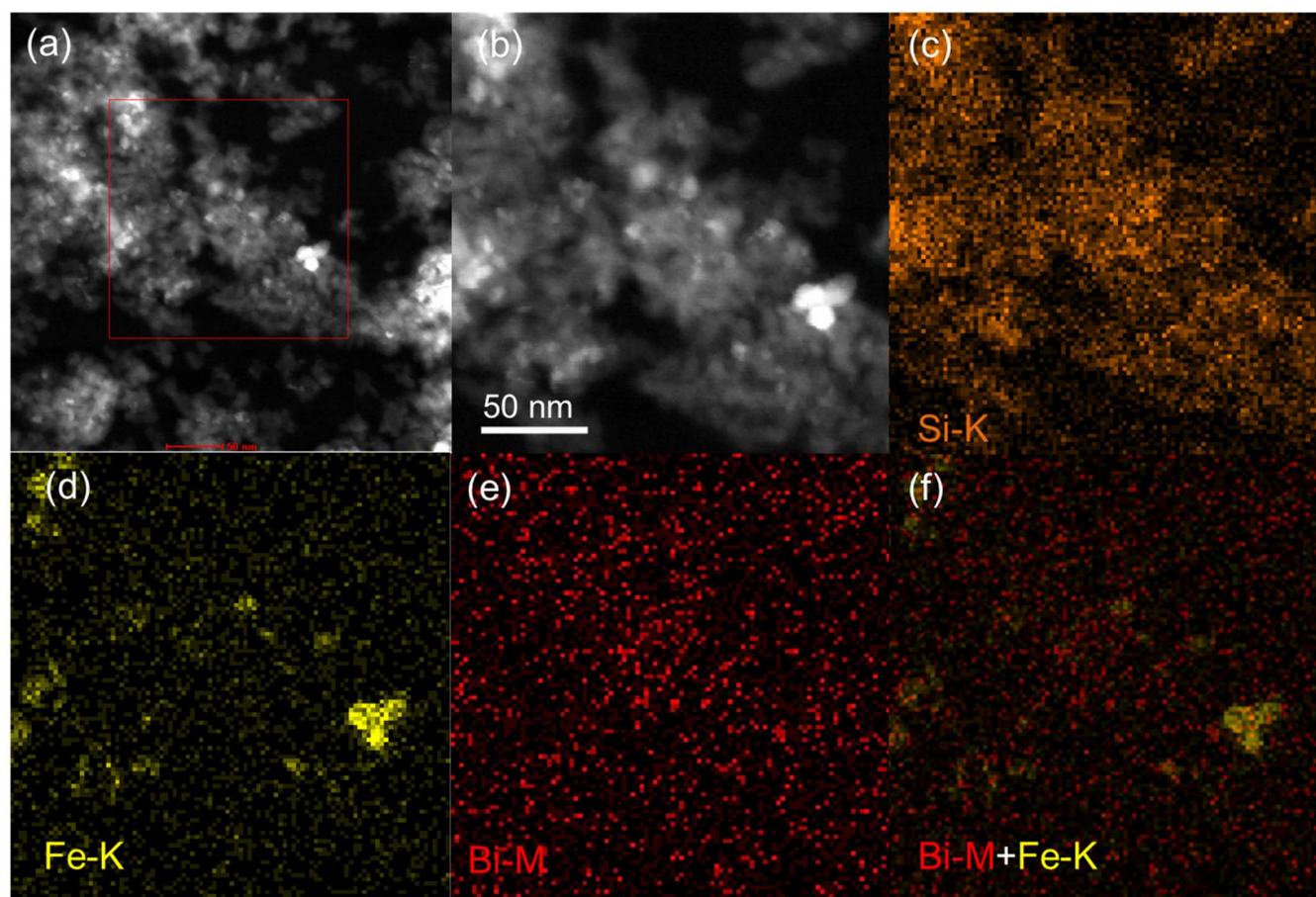


Figure S8. TEM image of Fe/SiO₂ + Bi/SiO₂ (a, b) before reaction with mapping of Si (c), Fe (d), Bi (e) and sum of Bi and Fe (f)

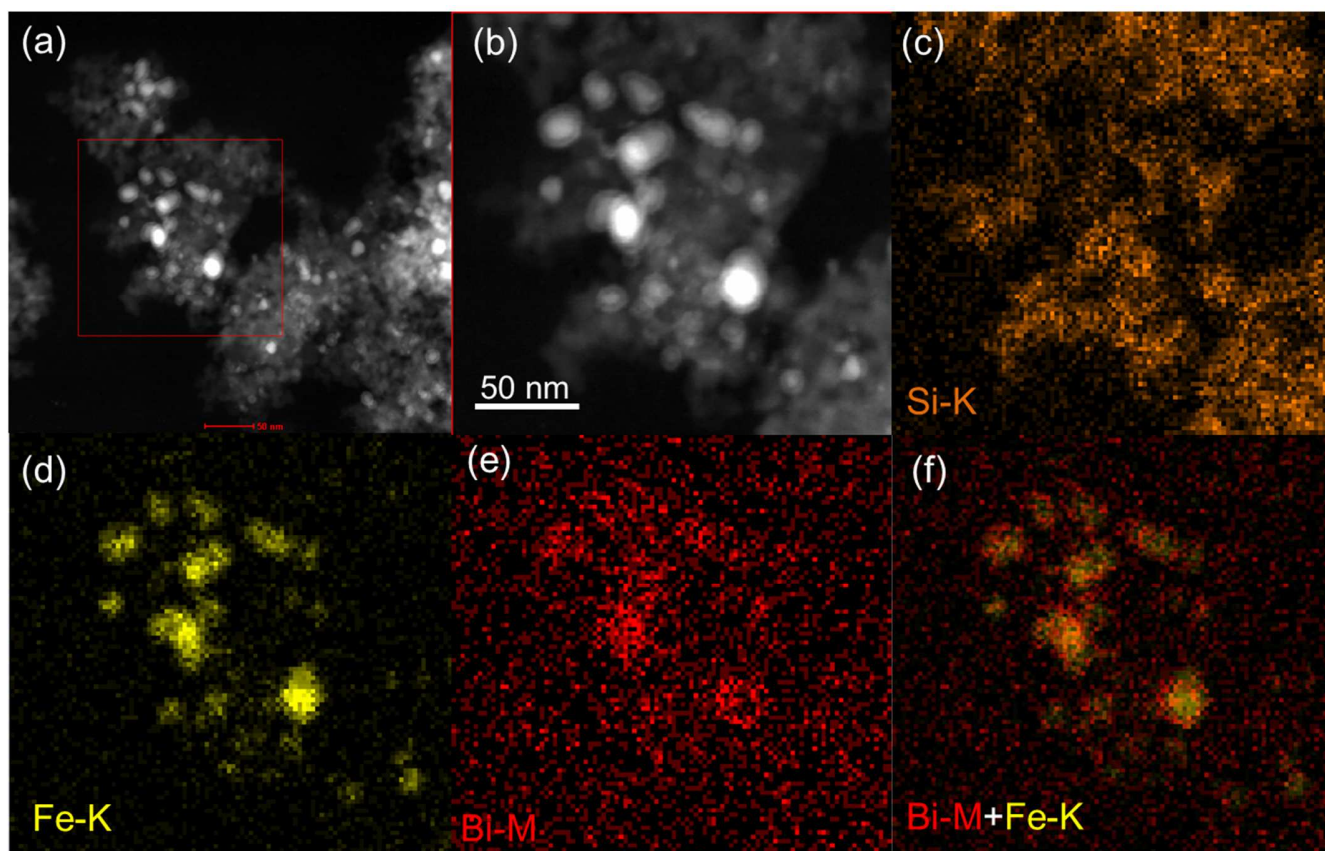


Figure S9. TEM image of Fe/SiO₂ + Bi/SiO₂ (a, b) after reaction with mapping of Si (c), Fe (d), Bi (e) and sum of Bi and Fe (f)

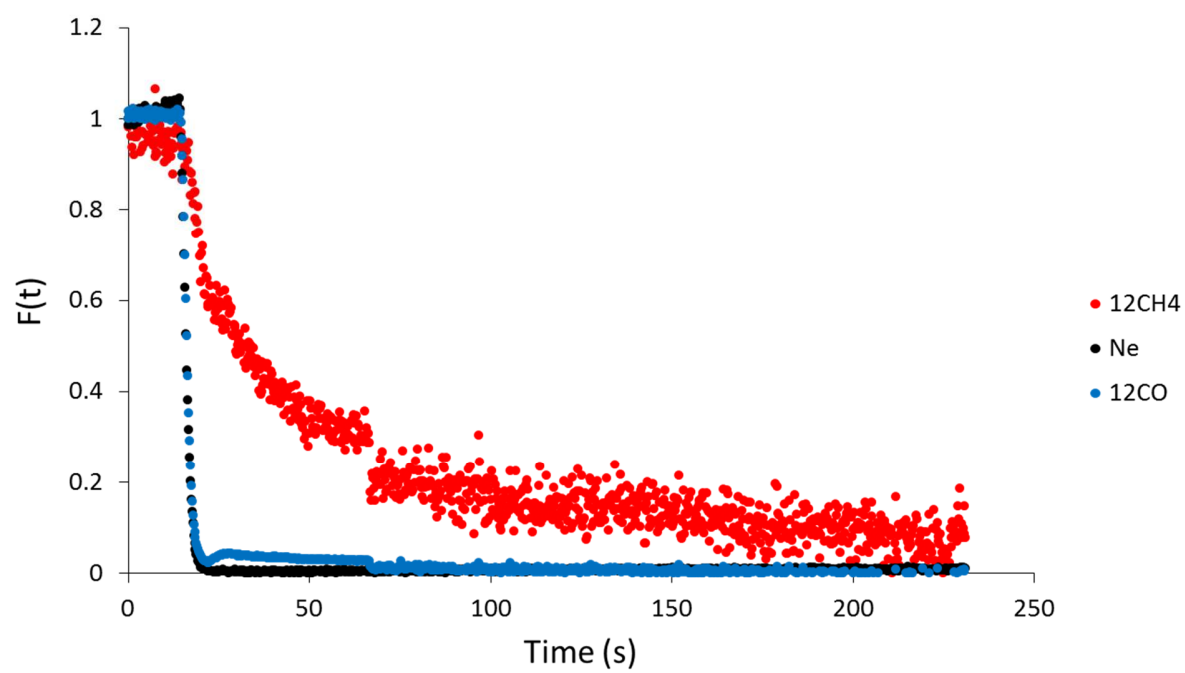


Figure S10. SSITKA results over FeBi/SiO₂. Reaction conditions: P= 1 bar, H₂/CO=2, GHSV =8.4 Lg⁻¹h⁻¹, T=350 °C.

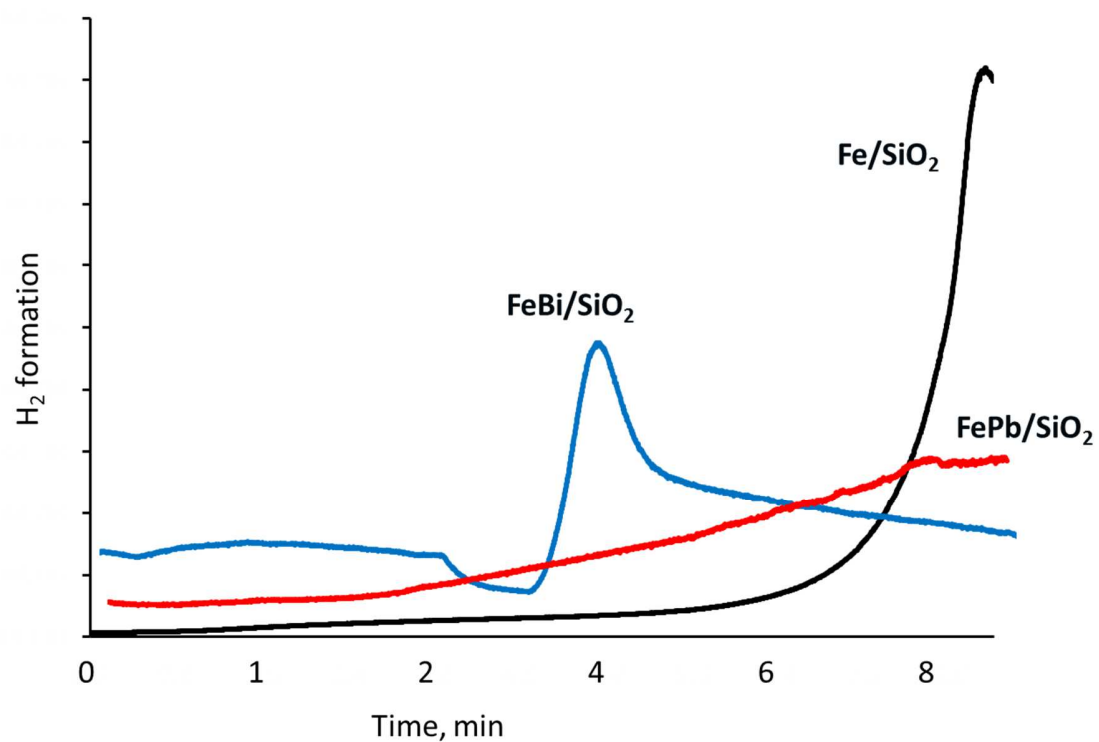


Figure S11. Rate of hydrogen production after exposure of the activated silica supported iron catalysts to water vapours. Reaction conditions: $P = 1$ bar, H_2O flow = 0.3 ml/h, $T = 350$ °C.